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EXAMINER	
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2441	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/532,696	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 12/07/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-22 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-22 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 02/17/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-22*** are pending.

2. ***Claims 1-22 are rejected.***

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claims **1-22** are rejected on the ground of nonstatutory double patenting as being unpatentable over claims **1-26** of U.S. Patent No. (not yet assigned (application number 17674281)). Although the claims at issue are not identical, they are not patentably distinct from each other because the instant claims states an external communication permission value and the patent claims states a mobile system instead of a vehicle. Since a mobile system can be part of a vehicle and a policy can include an external communication permission value, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify the patent in a way to match the instant claimed limitation.

Instant claims	Patent claims
1. A system comprising: - a vehicle having a first network zone and a second network zone of a different type than the first network zone; - a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising:	1. A system, comprising: - **a mobile system** having a first network zone and a second network zone of a different type than the first network zone; - a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising:

- a policy manager circuit structured to interpret a policy comprising a network regulation description and **an external communication permission value**;	- a policy manager circuit structured to interpret a policy comprising a network regulation description;
- a configuration circuit structured to configure **at least one** network interface circuit in response **to the policy** and to configure **at least one** gatekeeper interface circuit in response to the **policy**; **the at least one network interface circuit**, wherein **the at least one** network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone; and **the gatekeeper interface circuit,** wherein the gatekeeper interface circuit is configured to regulate communications between end points of at least one of the first network zone or the second network zone; and an external device.	- a configuration circuit structured to configure a first network interface circuit in response to the network regulation description, and to configure a gatekeeper interface circuit in response to the network regulation description; wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone; and wherein the gatekeeper interface circuit is structured to regulate communications between end points of the first network zone with at least one of an external communications portal or an external device

Claims **1-22** are rejected on the ground of nonstatutory double patenting as being unpatentable over claims **1-10** of US Patent No. 11,252,039. Although the claims at issue are not identical, they are not patentably distinct from each other because the instant claims have a broader scope than corresponding to the patent system claims and it would have been obvious to one ordinary skill in the

art before the effective filing date of the invention to omit the additional limitation in the patent claims to achieve the instant claimed limitation.

Instant claims	Patent claims
1. A system comprising: - a vehicle having a first network zone and a second network zone of a different type than the first network zone; - a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: - a policy manager circuit structured to interpret a policy comprising a network regulation description and **an external communication permission value**; - a configuration circuit structured to configure **at least one** network interface circuit in response **to the policy** and to configure **at least one** gatekeeper interface circuit in response to the **policy**; **the at least one network interface circuit**, wherein **the at least one** network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone; and	1. A system, comprising: - a vehicle having a first network zone and a second network zone of a different type than the first network zone; - a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: - a policy manager circuit structured to interpret a policy comprising a network regulation description; and - a configuration circuit structured to configure a first network interface circuit in response to the network regulation description, and to configure a gatekeeper interface circuit in response to the network regulation description, wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone, and

the gatekeeper interface circuit, wherein the gatekeeper interface circuit is configured to regulate communications between end points of at least one of the first network zone or the second network zone; and an external device.	wherein the gatekeeper interface circuit is structured to regulate communications between end points of the first network zone with at least one of an external communications portal or an external device.

Claim Interpretation

The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not

preceded by a structural modifier. Such claim limitation(s) is/are: "a policy manager circuit structured to interpret, a configuration circuit structured to configure, and a gatekeeper interface circuit structured to regulate" in claim 1. The corresponding structures and algorithms used to link to the above limitations are the structures and algorithms stated in the specification (for example paragraph 0101, 0152, 0291-0296, 0307-0310)

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1-8 and 20-22 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter

which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 1 recites the limitation " the gatekeeper interface circuit" in line 11 of the claim. There is insufficient antecedent basis for this limitation in the claim.

Claims 2-8 depends on claim 1 and are rejected under the same rationale.

Claim 20 recites the limitation " the gatekeeper interface circuit" in line 11 of the claim. There is insufficient antecedent basis for this limitation in the claim.

Claims 21-22 depends on claim 20 and are rejected under the same rationale.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

3. ***Claims 9-17 are rejected*** under 35 U.S.C. 102(a)(2) as being anticipated by BARRETT et al (Pub. No.: US 2019/0394089 A1).

As per claim 9, BARRETT discloses a method, comprising:

- interpreting a policy comprising a network regulation description and an external communication permission value (BARRETT, Fig 2, paragraph 0037, inter-domain firewalling can ensure that only certain types of traffic can pass from the external network 205 into the trusted

network 220 internal to the vehicle 202. An internal firewall, FW.sub.int, 230 is placed between the switch or router 210 of the the trusted network 220 and the gateway 240 of the IZ 225. An external firewall, FW.sub.ext, 250 is placed between the gateway 240 of the IZ 225 and the telecoms control unit 260);

- ***regulating communications between end points of a first network zone of a vehicle and end points of a second network zone of the vehicle in response to the network regulation description*** {BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear on a whitelist to pass through the firewall, while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report, and/or take other action);

- ***regulating communications, in response to the external communication permission value, between: end points of at least one of the first network zone or the second network zone*** {BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear on a whitelist to pass through the firewall, while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report, and/or take other action);

- ***wherein the second network zone is of a different type than the first network zone*** (BARRETT, Fig 1, 2, 3, paragraph 0021, 0027-0028 vehicle network, the internal network 100 can be a heterogeneous network. For example, a vehicle can integrate over 100 ECUs connected over multiple networks such as CAN (Control Area Network), LIN (Local Interconnect Network), FlexRay)*;*

As pre claim 10, claim 9 is incorporated and BARRETT discloses receiving a policy communication from the external device, and performing one of storing the policy or updating the policy in response to the policy communication (BARRETT, paragraph 0029, 0033, wherein software (including servers and clients) deployed in the IZ 225 can be tailored for vehicles. For example, the OTA software update service might act as a client towards an OTA server on an external network whilst acting as a server to clients within the trusted network that need the software updates. As another example, infotainment services can include applications that are commonly downloaded to handsets, but which are being deployed on the car infotainment system itself. In some implementations, some of the apps in the in-vehicle experience domain 232 obtained from the external network may be vehicle specific. Web browsing type functionality might also be provided on the infotainment system. In some implementations, the update is expected to be infrequent and the OEM can also provide/derive new firewall rules at the same time as the software update)*;*

As pre claim 11, claim 10 is incorporated and BARRETT discloses verifying the policy before performing the one of the storing the policy or updating the policy (BARRETT, paragraph 0093, wherein he firewall configurations can be verified by a verification test. For example, once the firewall configurations have been derived, they can be tested by deploying

them to one of the environments described above for dynamic analyses (e.g., either to the complete vehicle or to the test bench mounted system). The car software can be then executed according to one of the techniques described with respect to the dynamic analyses. Then, a process within the firewall system (e.g., the intrusion detection) can record whether any packets are blocked from getting through. The information regarding what packets were blocked (especially for example Layer 2, Layer 3, Layer 4, Application level information) can be returned to, for example, security engineers or other personnel who can take it into account in considering whether to provide a revised firewall configuration};

As pre claim 12, claim 11 is incorporated and BARRETT discloses **providing a notification to the external device in response to the verifying the policy cannot be applied** {BARRETT, paragraph 0093, wherein he firewall configurations can be verified by a verification test. For example, once the firewall configurations have been derived, they can be tested by deploying them to one of the environments described above for dynamic analyses (e.g., either to the complete vehicle or to the test bench mounted system). The car software can be then executed according to one of the techniques described with respect to the dynamic analyses. Then, a process within the firewall system (e.g., the intrusion detection) can record whether any packets are blocked from getting through. The information regarding what packets were blocked (especially for example Layer 2, Layer 3, Layer 4, Application level information) can be returned to, for example, security engineers or other personnel who can take it into account in considering whether to provide a revised firewall configuration};

As pre claim 13, claim 12 is incorporated and BARRETT discloses **wherein verifying the policy cannot be applied comprises at least one operation selected from the operations**

consisting of: determining that the policy violates a security criteria of a previously stored policy or the policy determining that the policy cannot be implemented due to a data storage limit; or determining that the policy cannot be implemented due to an external communication limit (BARRETT, paragraph 0093, wherein he firewall configurations can be verified by a verification test. For example, once the firewall configurations have been derived, they can be tested by deploying them to one of the environments described above for dynamic analyses (e.g., either to the complete vehicle or to the test bench mounted system). The car software can be then executed according to one of the techniques described with respect to the dynamic analyses. Then, a process within the firewall system (e.g., the intrusion detection) can record whether any packets are blocked from getting through. The information regarding what packets were blocked (especially for example Layer 2, Layer 3, Layer 4, Application level information) can be returned to, for example, security engineers or other personnel who can take it into account in considering whether to provide a revised firewall configuration);

As pre claim 14, claim 9 is incorporated and BARRETT discloses regulating external communications of end points of each of the first network zone and the second network zone in response to the policy (BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear on a whitelist to pass through the firewall, while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report, and/or take other action);

As pre claim 15, claim 9 is incorporated and BARRETT discloses regulating communications between a first end point of the first network zone and a second end point of the first network zone in response to the policy (BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear on a whitelist to pass through the firewall, while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report, and/or take other action);

As pre claim 16, claim 9 is incorporated and BARRETT discloses updating a local configuration file of a configurable ethernet switch in response to one of an implementation of the policy or an update of the policy (BARRETT, paragraph 0097 Upon receiving the updated software code, the example method 700 can go back to 704 for performing an analysis of the updated software code to identify connectivity of the network nodes in the vehicle network. The example method 700 can further proceed to 707 for determining updated connectivity information of the network nodes in the vehicle network based on the analysis; to 708 for updating configurations of the one or more firewalls in the vehicle network based on the updated connectivity information, and so on);

As pre claim 17, claim 9 is incorporated and BARRETT discloses updating a local configuration file of a configurable edge gateway in response to one of an implementation of the policy or an update of the policy (BARRETT, paragraph 0097 Upon receiving the updated software code, the example method 700 can go back to 704 for performing an analysis of the updated software code to identify connectivity of the network nodes in the vehicle network. The example method 700 can further proceed to 707 for determining updated connectivity information of the network nodes in the vehicle network based on the analysis; to 708 for updating configurations of the one or more firewalls in the vehicle network based on the updated connectivity information, and so on);

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441